

Lecture #8: Joint Distributions

MATH 321 – Mathematical Statistics

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Multivariate Random Variables

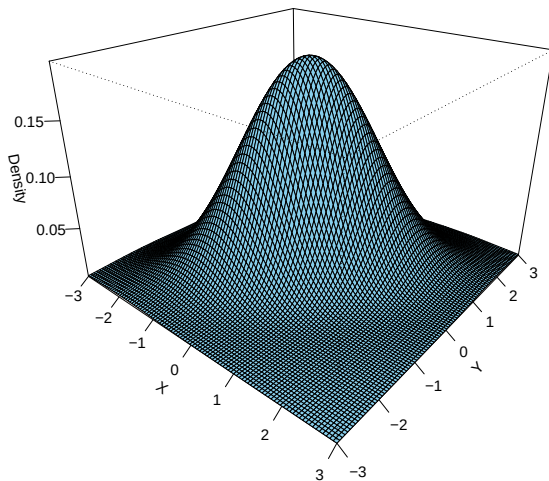
It's all well and good to consider random variables in isolation, but to do the actual interesting things you all want to do in probability and statistics, we need to understand how random variables relate to one another.

For instance, most of inferential statistics is built on the assumption of **indentically and independently distributed (i.i.d.) random variables**.

Similarly, linear regression is built on a specific relationship of random variables:

$$Y = \beta X + \varepsilon, \quad \varepsilon \sim N(0, 1)$$

Bivariate Normal Distribution



Joint Probability Mass Functions

Let X and Y be two discrete random variables. The **joint probability mass function (PMF)** of X and Y is defined as:

$$\begin{aligned} p_{X,Y}(x, y) &= P[\{X = x\} \cap \{Y = y\}] \\ &= P[(\omega, \xi) \mid \{X(\omega) = x\} \cap \{Y(\xi) = y\}]. \end{aligned}$$

If the area X and Y , A , are defined on is a subset of $\mathbb{R}^2$ ($A \subset \mathbb{R} \times \mathbb{R}$), then:

$$P(A) = \sum_{(x,y) \in A} p_{X,Y}(x, y).$$

Note that these are valid PMFs! All properties apply!

Joint PMF Example

Example: Let X be a coin toss and Y be a die roll. Find $p_{X,Y}$.

Y

1

2

3

4

5

6

X

H

$\frac{1}{12}$

$\frac{1}{12}$

1

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Joint Probability Density Functions

Let X and Y be two continuous random variables. The **joint probability density function (PDF)** of X and Y is a function $f_{X,Y}(x,y)$ that can be integrated to yield a probability:

$$P(\mathcal{A}) = \int_{\mathcal{A}} f_{X,Y}(x,y) dx dy,$$

for any event $\mathcal{A} \subseteq \Omega_X \times \Omega_Y$.

In particular, if $A = [a, b] \times [c, d]$, then

$$P(\mathcal{A}) = \iint_A f_{X,Y}(x,y) dA = \int_a^b \int_c^d f_{X,Y}(x,y) dy dx.$$

Note that these are valid PDFs! All properties apply!

Joint PDF Example

Example: Let X and Y be jointly uniformly distributed on $A = [1, 3] \times [1, 4]$.

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{6}, & \text{for } 1 \leq x \leq 3, 1 \leq y \leq 4, \\ 0, & \text{otherwise.} \end{cases}$$

- 1 What is the probability that $X \geq 2$ and $Y \geq 2$?

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- ❶ What is the probability that $X \geq 2$ and $Y \geq 2$?
- ❷ What is the probability that X and Y are less than 1 apart?

Marginal PMFs and Marginal PDFs

The **marginal probability mass function (PMF)** is defined as:

$$p_X(x) = \sum_{y \in \Omega_Y} p_{X,Y}(x, y) \quad \text{and} \quad p_Y(y) = \sum_{x \in \Omega_X} p_{X,Y}(x, y). \quad (1)$$

The **marginal probability density function (PDF)** is defined as:

$$f_X(x) = \int_{\Omega_Y} p_{X,Y}(x, y) dy \quad \text{and} \quad f_Y(y) = \int_{\Omega_X} p_{X,Y}(x, y) dx. \quad (2)$$

Note that these are valid PMFs/PDFs! All properties apply!

Joint PMF Example

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Y

1

2

3

4

5

6

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H

$\frac{1}{12}$

$\frac{1}{12}$

1

Joint PDF Example

Example: Let X and Y be jointly uniformly distributed on $A = [1, 3] \times [1, 4]$.

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{6}, & \text{for } 1 \leq x \leq 3, 1 \leq y \leq 4, \\ 0, & \text{otherwise.} \end{cases}$$

- ③ Find the marginal PDFs of X and Y .

Extending to N Random Variables

A collection of random variables $X_1, X_2, \dots, X_N$ is called **independent and identically distributed (i.i.d.)** if:

- 1 All $X_1, X_2, \dots, X_N$ are mutually independent.

Extending to N Random Variables

A collection of random variables $X_1, X_2, \dots, X_N$ is called **independent and identically distributed (i.i.d.)** if:

- 1 All $X_1, X_2, \dots, X_N$ are mutually independent.
- 2 All $X_1, X_2, \dots, X_N$ have the same distribution, i.e. $f_{X_1}(x) = f_{X_2}(x) = \dots = f_{X_N}(x)$ for all x .

Joint Expectation

Let X and Y be two random variables. The **joint expectation** is

$$E[XY] = \sum_{y \in \Omega_Y} \sum_{x \in \Omega_X} xy p_{X,Y}(x, y),$$

if X and Y are discrete, or

$$E[XY] = \int_{\Omega_Y} \int_{\Omega_X} xy f_{X,Y}(x, y) dx dy,$$

if X and Y are continuous.

Example: Suppose $X, Y \sim N(0, 1)$. Find $E(XY)$.

Let X and Y be two random variables. Then the **covariance** of X and Y is

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)],$$

where $\mu_X = E[X]$ and $\mu_Y = E[Y]$.

Note that $\text{Cov}(X, X) = \text{Var}(X)$.

Theorem: Let X and Y be two random variables. Then

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

If $X_1, X_2, \dots, X_n$ are random variable, then $\text{Cov}(X_i, X_j)$ makes the **covariance matrix**, Σ .

Theorem: For any X and Y ,

- ① $E[X + Y] = E[X] + E[Y]$.
- ② $Var[X + Y] = Var[X] + Var[Y] + 2Cov(X, Y)$.

Covariance and Independence

Theorem: If X and Y are independent, then

$$E[XY] = E[X]E[Y].$$

Theorem: Consider the following two statements:

- a. X and Y are independent.
 - b. $Cov(X, Y) = 0$.
- 1 Does statement (a) imply statement (b)?

Covariance and Independence

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$$E[XY] = E[X]E[Y].$$

Theorem: Consider the following two statements:

- a. X and Y are independent.
 - b. $Cov(X, Y) = 0$.
-
- 1 Does statement (a) imply statement (b)?
 - 2 Does statement (b) imply statement (a)?

Example of Zero Covariance but Non-Independent

Example: Consider the following two variables X and Y .

k

1

2

3

4

X

1

0

-1

0

Y

0

Example of Zero Covariance but Non-Independent

Example: Consider the following two variables X and Y .

k

1

2

3

4

X

1

0

-1

0

Y

0

The Correlation Coefficient

Let X and Y be two random variables. The **correlation coefficient** is

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}.$$

Properties:

- ① ρ is always between -1 and 1 , i.e. $-1 \leq \rho \leq 1$.
- ② When $X = Y$, $\rho = 1$.
- ③ When $X = -Y$, $\rho = -1$.
- ④ When X and Y are uncorrelated, $\rho = 0$.

Cauchy-Schwarz Inequality

Theorem: For any random variables X and Y ,

$$(E[XY])^2 \leq E[X^2]E[Y^2].$$